Project – 2 Waste-Heat-Driven Ammonia–Water Absorption Chiller for a District Cooling Network

2.1 Background

Industrial sites reject large quantities of medium-grade heat in the form of flue gas at 250–350 °C which is too cool for power generation but ideal for driving a thermally activated chiller. A ammonia–water absorption refrigeration unit converts that heat into chilled water, which a district cooling network distributes to air handling units (AHUs) in the surrounding buildings. Unlike a vapour-compression plant, its capacity and heat demand are strongly coupled to the ambient condition through the condenser and absorber, so the same machine behaves very differently across an Indian summer, monsoon, and winter. Your task is to model the full chain: outdoor air → AHU psychrometrics → chilled-water load → absorption cycle → waste-heat demand and size the components that must survive the whole year.

2.2 System Description

A. Waste heat source

- Process flue gas, treated as an ideal gas with constant C$p = 1.10$ kJ/kg·K, available at 300 °C, up to a maximum of 80 kg/s.
- Flue gas must leave the generator heat exchanger at ≥ 140 °C.
- Generator heat exchanger effectiveness $\varepsilon_{gen} = 0.75$, defined on the flue-gas side: $\varepsilon_{gen} = (T_{fg,in} - T_{fg,out})/(T_{fg,in} - T_{gen})$, where T_{gen} is the solution temperature leaving the generator.

The "heating load" asked for in Task 1 is the generator heat duty Q_{gen}; the flue-gas mass flow that supplies it follows from the effectiveness and the 140 °C floor.

B. Ammonia-water chiller unit

- Components: generator with rectifier (vapour leaving at 99.5 % NH_3 by mass), condenser, expansion valve, evaporator, absorber, solution pump $(\eta_{pump} = 0.70)$, solution heat exchanger $(\varepsilon_{SHX} = 0.70)$, weak-solution throttle. A refrigerant precooler is optional (state clearly if used).
- Evaporator: $T_{evap} = 2$ °C, supplying chilled water at 7 °C with 12 °C return.
- Condenser and absorber are cooled by the same medium; assume $T_{abs} = T_{cond}$.
- Generator outlet (weak solution), absorber outlet (strong solution), condenser outlet (saturated liquid), and evaporator outlet (saturated vapour) are all saturated states at their respective pressures.
- Pressure drops in vessels and piping are neglected; $P_{high} = P_{sat,NH_3(T_{cond})}$, $P_{low} = P_{sat,NH_3(T_{evap})}$.
- Minimum stable turndown of the chiller: 25 % of design capacity

Property models: No black-box property libraries for ammonia or moist air. CoolProp is permitted only for pure water/steam (condensate, chilled water, any steam-side calculation, and the saturation pressure of water used inside the psychrometric relations below). Ammonia and moist air must be evaluated using thermodynamic relations as discussed in class lectures with a real-gas equation of state used wherever the fluid's compressibility departs materially from unity.

- Reference state (for enthalpy and entropy estimation): $T_{ref} = 273.15$ K, $P_{ref} = 101.325$ kPa (standard pressure and temperature), ideal-gas state, $h_{ref} = 0, s_{ref} = 0$. State this explicitly in the report and carry it through consistently.
- Ammonia–water solution: Pátek & Klomfar (1995) (https://doi.org/10.1016/0140-7007(95)00006-W) correlations for bubble/dew temperature and liquid/vapour enthalpy as functions of (P, x). Implement these yourself; do not use a pre-packaged absorption-cycle library.

Pátek & Klomfar Model

Bubble point (saturated liquid at pressure p, liquid mole fraction x)

$$T_{bub}(p,x) = T_0 \sum_i a_i (1-x)^{m_i} \left[\ln\left(\frac{p_o}{p}\right)\right]^{n_i}$$

Table 1: Exponents and coefficients

i	m_i	n_i	a_i
1	0	0	3.223020E+00
2	0	1	-3.842060E-01
3	0	2	4.609650E-02
4	0	3	-3.789450E-03
5	0	4	1.356100E-04
6	1	0	4.877550E-01
7	1	1	-1.201080E-01
8	1	2	1.061540E-02
9	2	3	-5.335890E-04
10	4	0	7.850410E+00
11	5	0	-1.159410E+01
12	5	1	-5.231500E-02
13	6	0	4.895960E-02
14	13	1	4.210590E-02

$T_o = 100K$ and $p_o = 2MPa$

Dew point (saturated vapor at pressure p, vapor mole fraction y)

$$T_{dew}(p,y) = T_0 \sum_i a_i (1-y)^{\frac{m_i}{4}} \left[\ln\left(\frac{p_o}{p}\right)\right]^{n_i}$$

Table 2: Exponents and coefficients

i	m_i	n_i	a_i
1	0	0	3.240040E+00
2	0	1	-3.959200E-01
3	0	2	-4.356240E-02
4	0	3	-2.189430E-03
5	1	0	-1.435260E+00
6	1	1	1.052560E+00
7	1	2	-7.192810E-02
8	2	0	1.223620E+01
9	2	1	-2.243680E+00

10	3	0	-2.107800E+01
11	3	1	1.108340E+00
12	4	0	1.453990E+01
13	4	2	6.443120E-01
14	5	0	-2.212640E+00
15	5	2	-7.562660E-01
16	6	0	-1.355290E+00
17	7	2	1.835410E-01

$T_o = 100K$ and $p_o = 2MPa$

- Pure ammonia (refrigerant side): a cubic real-gas equation of state: Peng–Robinson or Redlich–Kwong–Soave using $T_c = 405.56$ K, $P_c = 11.333$ MPa, $\omega = 0.256$, $MW = 17.031$ kg/kmol. Compute $h(T,P)$ and $s(T,P)$ as an ideal-gas contribution (integrate the Cp_0 polynomial in Table A.6 from T_{ref}) plus the EOS departure function at the actual (T, P); the saturation pressure $P_{sat}(T)$ is not read off a table but obtained from a generalized vapour-pressure correlation (e.g. the Lee–Kesler/Pitzer acentric-factor correlation, using the same T_c, P_c, ω). State clearly why a real-gas EOS is appropriate here: at generator/condenser pressures (1–2 MPa, close to saturation) the compressibility factor of ammonia vapour departs several percent from 1, and the liquid phase is not ideal-gas at all.

  Lee-Kesler equation of state for simple fluids

$$Z = \frac{P_r v_r}{T_r} = a + \frac{B}{v_r} + \frac{C}{v_r^2} + \frac{D}{v_r^5} + \frac{c_4}{T_r^3 v_r^2}\left(\beta + \frac{\gamma}{v_r^2}\right)\exp\left(-\frac{\gamma}{v_r^2}\right)$$

$$B = b_1 - \frac{c_2}{T_r} - \frac{b_3}{T_r^2} - \frac{b_4}{T_r^3}$$

$$C = c_1 - \frac{c_2}{T_r} + \frac{c_3}{T_r^3}$$

$$D = d_1 + \frac{d_2}{T_r}$$

  Where

$$T_r = \frac{T}{Tc}, P_r = \frac{P}{P_c}, v_r = \frac{v}{\frac{RT_c}{P_c}}$$

Constant	Simple Fluids	Constant	Simple Fluids
b_1	0.1181193	c_3	0.0
b_2	0.265728	c_4	0.042724
b_3	0.154790	$d_1 \times 10^4$	0.155488
b_4	0.030323	$d_2 \times 10^4$	0.623689
c_1	0.0236744	β	0.65392
c_2	0.0186984	γ	0.060167

- Moist air: dry air and water vapour treated with the ideal-gas mixture relations ($w = 0.622P_v/(P - P_v), h = C_{p,da}T + wh_v(T)$), because at the pressures and temperatures of this problem (≈101 kPa, −5 to 45 °C) both dry air and water vapour have Z ≈ 0.999. The one place a real fluid correlation is required is the water-vapour saturation pressure $P_{sat,H2O(T)}$, which will come from Antoine's equation. The vapour enthalpy $h_v(T)$ at 1 atm may be taken as h_g of saturated water vapour at that T from CoolProp, consistent with the same 0 °C reference.

C. District cooling network and zones

- 400 identical zones, each 40 m^2 floor × 3 m height, served by chilled water at 7 °C supply / 12 °C return. Distribution heat gain = 5 % of delivered cooling; chilled-water pump heat = 2 %.
- Zone envelope: overall $UA_{env} = 80$ W/K (walls, roof, glazing combined). Solar gain 1200 W during the day, 0 at night.
- Internal gains: 4 occupants (75 W sensible + 55 W latent each) and 20 W/m^2 lighting + equipment (sensible).
- AHU: supply air $\dot{m}_{da} = 0.40$ kg/s dry air per zone, made up of 70 % recirculated room air and 30 % fresh outdoor air (by dry-air mass). Cooling coil bypass factor BF = 0.15. Fan heat neglected.
- Room set-point: dry-bulb held exactly at 22 °C by modulating the coil. Relative humidity must not exceed 75 %; the actual room humidity ratio is a result of the moisture balance, not an input, and must be iterated to convergence.
- Condensate from every coil falls into a drain pan, passes through a trap, drains by gravity to a building riser (20 zones per riser), and is collected in a central decanter/holding tank for reuse

D. Ambient Conditions

Condition	T_{db} (°C)	RH (%)	Solar
Summer day	40	50	on
Summer night	30	50	off
Monsoon day	35	85	on
Monsoon night	30	85	off
Winter day	28	50	on
Winter night	16	50	off

For annual quantities assume each season lasts four months, with 12 h day / 12 h night. Atmospheric pressure 101.325 kPa

2.3 Tasks

Task – 1 Cooling and heating loads; and ammonia circuit size across the seasons

- For each of the six conditions, solve the zone psychrometrics: outdoor state → adiabatic mixing (70/30) → coil (ADP and bypass factor) → supply state → room sensible and latent balances. The coil must deliver exactly the supply temperature required to hold 22 °C; the supply humidity ratio follows from the coil process; the room humidity ratio is then found by iterating the moisture balance until it converges. Report T_{mix}, ADP, w_{supply}, the resulting room RH, and flag any condition where the 75 % ceiling is violated and state what control action fixes it.

  ADP (Apparatus Dew Point) is the temperature a cooling coil would drive the air to if the coil had infinite contact area, i.e., if every bit of air actually touched the cold fins/tubes long enough to reach the coil's surface temperature. At that point the air would be saturated (100% RH) at the coil's effective surface temperature. It's called a "dew point" because it's the temperature at which the air's moisture just starts condensing out on the coil.

  Bypass Factor (BF) accounts for the fact that a real coil doesn't have infinite area — some fraction of the air slips through the coil without ever touching a cold surface, "bypassing" it. So the air leaving the coil is really a mix of:

  - *air that fully contacted the coil and left at the ADP (saturated), and*
  - *air that bypassed the coil entirely and left unchanged at the entering (mixed) condition.*

  If BF = 0.15, that means 15% of the air effectively bypasses the coil untouched, and 85% is brought all the way to the ADP

$$\mathrm{BF} = \frac{T_{supply} - ADP}{T_{mix} - ADP}$$

$$w_{supply} = w_{ADP} + BF(w_{mix} - w_{ADP})$$

$$T_{supply} = ADP + BF(T_{mix} - ADP)$$

- Compute the coil load per zone (with the condensate enthalpy term), scale to the district with distribution and pump gains, and obtain the chiller evaporator duty Q_{evap} for each condition. Identify the governing condition and explain why it is not the one with the highest dry-bulb temperature.
- Solve the absorption cycle at each condition with T_{cond} from your Task 2 condenser (first pass: $T_{cond} = T_{db} + 12$ K, then close the loop). Report Q_{gen} (the heating load), COP, refrigerant mass flow $\dot{m}_r$, strong- and weak-solution concentrations and flows, circulation ratio, solution-pump volumetric flow, and the required flue-gas mass flow. Check that the flue-gas flow never exceeds 80 kg/s and that the chiller never falls below 25 % turndown; flag it where it does, propose possible remedy (thermal storage, chiller staging, or free cooling).
- "Ammonia circuit size" means: $\dot{m}_r$, the solution pump duty (m^3/h and kPa head generation), and the vapour volumetric flow leaving the evaporator must be estimated

Task – 2 Condenser design for the full ambient spectrum

- Compare two condenser types: (i) air-cooled finned-tube (approach to dry-bulb; $U_{overall} = 30$ W/m^2·K on the finned-side area) and (ii) evaporative (approach to wet-bulb; $U_{overall} = 600$ W/m^2·K on wetted tube area, cycles of concentration 4). Compute the wet-bulb temperature of each condition from your own ideal-gas psychrometric relations (iterate the adiabatic-saturation energy balance to T_{wb} using CoolProp only for P_{sat,H_2O} and h_{g,H_2O}) and identify which condition governs each type.
- For your chosen type, fix the design point at the governing condition with a stated approach (e.g., 12 K to dry-bulb or 8 K to wet-bulb), compute Q_{cond} from the cycle, the LMTD, the required UA, the heat-transfer area, and the air or water flow.
- Off-design: holding UA fixed, solve for the condensing temperature at the other five conditions by iterating between the cycle model and the condenser energy balance. Show the resulting T_{cond}, P_{high}, generator temperature and COP for all six conditions.
- Verify at every condition that (a) $P_{high} \leq 20$ bar (vessel design pressure), (b) T_{gen} required at that P_high is deliverable by 300 °C flue gas with $\varepsilon_{gen} = 0.75$ and the 140 °C floor, and (c) $T_{cond} \geq 25$ °C so the expansion valve retains adequate pressure differential. Flag where (c) fails at low ambient

Task – 3 Water decanter (condensate) system

Condensate rate per zone and for the district at every condition (kg/h and m^3/day). Identify the governing condition for the decanter and explain why it differs from the chiller's governing condition.

Task – 4 Sensitivity to fresh-air fraction: 30 % → 40 %

- Re-run Tasks 1 and 3 with 40 % fresh air. Tabulate, for each condition, the percentage change in coil load, $\dot{Q}_{evap}$, $\dot{Q}_{gen}$, $\dot{m}_r$, flue-gas flow and condensate rate.
- Explain physically why the monsoon conditions are the most sensitive, and why the winter-night change is small.
- Check whether the 75 % RH ceiling still holds at monsoon night with 40 % fresh air. If it does not, quantify the lower ADP (and any reheat) needed and the resulting extra generator heat.

- State whether the 80 kg/s flue-gas limit or the 20 bar vessel limit becomes active anywhere with 40 % fresh air

2.4 Required Validation Test Case

Before any design results are accepted, each group must show that their code reproduces the following hand-traceable checks. The three checks are independent so that a discrepancy points to an error in the solver.

Check – 1 Psychrometric mixing (moist-air routine)

Given: 70 % room air at 22 °C, 75 % RH mixed adiabatically with 30 % outdoor air at 40 °C, 50 % RH, $P = 101.325$ kPa. Hand values use P_{sat,H_2O} from CoolProp (matches Table B.1.1 to 4 figures: 40 °C exact; 22 °C ≈ 2.645 kPa), $w = 0.622P_v/(P - P_v)$ and $h = 1.006T + w(2501 + 1.86T)$, consistent with the 0 °C ideal-gas reference.

State	P_v (kPa)	w (kg/kg da)	h (kJ/kg da)	T (°C)
Room, 22 °C/75% RH	1.984	0.01242	53.71	22
Outdoor, 40 °C/50% RH	3.692	0.02352	100.82	40
Mixed (0.7/0.3)		0.01575	67.84	27.48

Match these to within 0.5 % for w and h and 0.1 K for T. The tolerance is tight because ideal-gas mixing at this pressure should have no model error; a larger gap is a units or root-finding bug, not "real-gas correction." Also report T_{wb} for the summer-day and monsoon-day outdoor states (expected around 30.3 °C and 32.6 °C from the adiabatic-saturation iteration).

Check – 2 Ammonia real-gas EOS vs. Table B.2.1

Given: saturated ammonia at 0 °C and 40 °C (chosen to land on tabulated rows). Compute $P_{sat}(T)$ from your generalized vapour-pressure correlation and $h_{fg}(T)$ from the EOS departure functions. P_{sat} and h_{fg} are reference-independent, so they can be checked directly against the table; h_f and h_g depend on the 0 °C/101.325 kPa ideal-gas reference that was given in Section B and are not expected to equal the table's own (different) reference.

Quantity	0 °C	40 °C	Table B.2.1
P_{sat} (kPa)	424–430	1550–1570	429.6 / 1554.9
h_{fg} (kJ/kg)	1260–1300	1090–1150	1261.8 / 1098.8

A Peng–Robinson EOS with the Lee–Kesler/Pitzer vapour-pressure correlation and the critical constants above reproduces P_{sat} to within about 1.2 % and h_{fg} to within about 3–4 % of the table. This is the genuine, expected accuracy of a cubic EOS with generalized correlations for a polar, hydrogen-bonding fluid like ammonia, not numerical error. Report your own percentage deviation and discuss it: it is a legitimate finding, not a bug, provided it falls within roughly ±5 % on P_{sat} and ±8 % on h_{fg}. A deviation far outside that band (wrong sign, order of magnitude, or a value that doesn't decrease appropriately with T) does indicate an implementation error.

- Refrigerating effect: $q_e = h_{g(0\,°C,evap)} - h_{f(40\,°C,cond)}$. Because both terms share the same reference, the difference q_e must still match h_{fg}-based expectations: $q_e \approx 1050 - 1100$ kJ/kg.
- For $Q_{evap} = 5000$ kW: expected $\dot{m}_r \approx 4.5 - 4.8$ kg/s. Report your own value and compare.

Check – 3 Absorption-cycle balances (no external tables needed)

At any operating point the code produces, the following must hold:

$$Q_{gen} + Q_{evap} + W_{pump} = Q_{cond} + Q_{abs} + Q_{rect}$$

$$\dot{m}_{strong} x_{strong} = \dot{m}_{weak} x_{weak} + \dot{m}_r x_r$$

$$COP < COP_{rev} = \left[\frac{T_{evap}}{T_{cond} - T_{evap}}\right]\left[\frac{T_{gen} - T_{abs}}{T_{gen}}\right]$$

For $T_{evap} = 2$ °C, $T_{cond} = T_{abs} = 45$ °C, $T_{gen} = 130$ °C the reversible bound is ≈ 1.35; the actual COP of a single-effect NH_3–H_2O machine is expected to be range between 0.45–0.65. A COP outside this band, or one that rises with condensing temperature, indicates a sign or property-reference error.

Reporting requirement

Each report must include a Validation Table with columns Quantity | Hand value | Code output | % difference | Pass/Fail covering every quantity in Checks 1–3, plus a test_validation.py in the repository that asserts the same tolerances automatically.

2.5 Deliverables

Python repository with README of all the codes, requirements, the Pátek–Klomfar implementation as a separate tested module, and a single reproducible driver script.

A ppt giving the following:

#	Section	Required content
1	Schematic	Two labelled diagrams: (a) the absorption cycle with all state points numbered — generator, rectifier, condenser, valve, evaporator, absorber, pump, SHX, weak-solution throttle — and the flue-gas and cooling-medium streams; (b) one zone AHU showing return, fresh-air, mixing, coil, drain pan/trap and supply, with the psychrometric state points numbered. Both numbering schemes must be used consistently in every equation and table that follows.
2	Algorithm	Flowchart or structured pseudocode: the room-humidity iteration, the coil ADP/bypass solution, the Pátek–Klomfar property calls, the cycle solution sequence.
3	Analysis	Validation table; psychrometric chart with all six mixing/coil processes drawn; seasonal tables for Tasks 1–3; condenser type comparison and off-design map; the 40 % fresh-air sensitivity (Task 4).
4	Conclusion	One page: the governing condition for each component (chiller, condenser, decanter), the recommended condenser type and size, the flue-gas demand envelope over the year, the response to 40 % fresh air, and the single biggest design lever.

2.6 Grading Rubric

Criterion	Weight
Psychrometric and cycle model fidelity (own ideal-gas moist-air relations, own ammonia real-gas EOS, own Pátek–Klomfar, correct room-humidity iteration, correct condensate enthalpy term)	25
Validation (all 3 checks within tolerance, reference-state shift shown)	20
Seasonal load/heating-load/circuit-size results and governing-condition reasoning (Task 1)	15
Condenser design and off-design map with all three feasibility checks (Task 2)	10
Decanter sizing and fresh-air sensitivity (Tasks 3–4)	10
Report and presentation	20